V2X-Sim — Cooperative Perception Study

Configuration G Ablation Report — v2 (post-fix)

Three-arm V2X / V2I / V2V penetration sweep, rerun after the braking-logic fixes

At a glance
Second iteration of Configuration G, run after fixing the v1 braking regression. 72/72 cells completed with no gaps (one transient CARLA crash self-recovered on retry). The fix is verified: pooled collisions fell vs v1 and the cooperative-braking layer now actively engages. **Honest caveat:** even post-fix the penetration curve is not a clean monotonic safety gain — collisions still rise from the all-human baseline to any connected condition and plateau, and the three arms stay close. Single seed; dense Town10 dominates. Directional, not conclusive.

1. Experimental design

Dimension	Values
Communication arms	Combined (V2X+V2V) | V2I-only (--no-v2v) | V2V-only (--no-v2x)
Penetration p	0.0, 0.5, 0.9, 1.0
Maps	Town01 (grid), Town05 (multi-lane), Town10HD_Opt (dense downtown)
Weather	ClearNoon, HardRainNoon
Seeds	0 (single seed)
Sim duration	60 s/cell @ dt 0.05 (20 Hz)
Population	60 vehicles + 60 walkers
HDV / CAV	HOSTILE_MIX (30% aggressive, 40% ignore-veh) / AI_REALISTIC (3 m gap, ~1% slips)
CPM / V2V cadence	10 Hz (100 ms)
Cells per arm / total	24 / 72

Metric note: primary vs total collisions
coll = total deduplicated incidents (one per vehicle pair). **primary** excludes *secondary* incidents - a new pair where one party was already immobilised by an earlier crash (traffic piling into a stationary wreck, not a fresh failure). Primary is the cleaner safety signal because immobilise-in-place can otherwise let one wreck, bumped by several cars, count several times.

2. Results — collisions by penetration and arm

2a. Total collisions (avg over maps x weather)

p	Combined	V2I-only	V2V-only
0.0	5.8	6.5	6.3
0.5	8.5	8.3	9.2
0.9	9.5	7.7	9.0
1.0	9.2	8.7	9.2

2b. Primary collisions (pileups into frozen wrecks excluded)

p	Combined	V2I-only	V2V-only
0.0	5.7	5.8	5.8

0.5	7.8	7.7	7.8
0.9	8.0	6.8	8.0
1.0	7.7	7.2	7.8

Collisions still rise from the all-human baseline (~6) to any connected condition (~7-8 primary) and then plateau; the three arms remain close. With a single seed and the dense hostile Town10HD_Opt dominating the average, treat this as directional.

2c. Who is crashing (Combined arm)

p	CAV	HDV	VRU (pedestrian)
0.0	0.0	5.8	1.0
0.5	6.8	1.7	0.7
0.9	9.5	0.0	0.0
1.0	9.2	0.0	0.0

3. Collisions by map

3a. Total (avg over weather)

Map	Arm	p0.0	p0.5	p0.9	p1.0
Town01	Combined (V2X+V2V)	5.0	6.0	9.0	6.5
Town01	V2I-only (no V2V)	5.5	8.0	5.0	8.5
Town01	V2V-only (no V2X)	4.0	9.5	8.0	6.0
Town05	Combined (V2X+V2V)	4.0	6.5	7.5	8.5
Town05	V2I-only (no V2V)	2.5	6.0	6.5	5.5
Town05	V2V-only (no V2X)	4.0	4.5	6.0	6.5
Town10HD_Opt	Combined (V2X+V2V)	8.5	13.0	12.0	12.5
Town10HD_Opt	V2I-only (no V2V)	11.5	11.0	11.5	12.0
Town10HD_Opt	V2V-only (no V2X)	11.0	13.5	13.0	15.0

3b. Primary (avg over weather)

Map	Arm	p0.0	p0.5	p0.9	p1.0
Town01	Combined (V2X+V2V)	5.0	6.0	7.5	5.0
Town01	V2I-only (no V2V)	5.5	8.0	4.5	7.5
Town01	V2V-only (no V2X)	4.0	9.0	6.5	5.5
Town05	Combined (V2X+V2V)	4.0	6.0	6.0	7.5
Town05	V2I-only (no V2V)	2.5	5.5	5.5	5.0
Town05	V2V-only (no V2X)	4.0	4.5	6.0	6.5
Town10HD_Opt	Combined (V2X+V2V)	8.0	11.5	10.5	10.5
Town10HD_Opt	V2I-only (no V2V)	9.5	9.5	10.5	9.0
Town10HD_Opt	V2V-only (no V2X)	9.5	10.0	11.5	11.5

Town10HD_Opt carries most of the absolute load and most of the secondary pileups; Town01/Town05 are milder.

4. Collisions by weather (avg over maps, total)

Weather	Arm	p0.0	p0.5	p0.9	p1.0
ClearNoon	Combined (V2X+V2V)	6.0	8.0	10.3	8.7
ClearNoon	V2I-only (no V2V)	6.7	8.0	7.3	8.0
ClearNoon	V2V-only (no V2X)	6.3	8.7	9.3	10.0
HardRainNoon	Combined (V2X+V2V)	5.7	9.0	8.7	9.7
HardRainNoon	V2I-only (no V2V)	6.3	8.7	8.0	9.3
HardRainNoon	V2V-only (no V2X)	6.3	9.7	8.7	8.3

5. Effect of the fixes — v2 (post-fix) vs v1 (pre-fix)

p	v1 total	v2 total	v2 primary	change
0.0	5.8	6.2	5.8	+0.4
0.5	10.0	8.7	7.8	-1.3

0.9	9.5	8.7	7.6	-0.7
1.0	9.8	9.0	7.6	-0.8

Cooperative + emergency braking now fire (Combined arm, avg per cell):

p	v1 coop	v2 coop	v1 hard	v2 hard
0.0	0	0	0	0
0.5	11	187	432	1276
0.9	33	362	811	1321
1.0	65	385	587	1442

The velocity-matched ego-ghost filter un-blinded CAVs to close-range threats; they now brake when they should, and those hard brakes propagate as V2V brake-intent so the cooperative layer engages instead of sitting idle.

6. Cooperative-communication load

p	V2V recv (comb.)	Coop (comb.)	Coop (v2v-only)	Coop (v2i-only)
0.0	0	0	0	0
0.5	116846	187	43	0
0.9	410705	362	163	0
1.0	493772	385	188	0

v2i-only correctly shows zero cooperative brakes (no V2V); V2V load scales with penetration.

7. Performance (wall/sim, Combined arm)

Map	p0.0	p0.5	p0.9	p1.0
Town01	0.58	0.85	1.34	1.62
Town05	0.61	0.94	1.85	2.10
Town10HD_Opt	0.67	1.46	3.37	4.04

8. Findings

- **The v1 regression is fixed.** Replacing the position-only 3.0 m ego self-exclusion with a velocity-matched ego-ghost filter restored close-range emergency braking; pooled p=1.0 collisions fell and Combined-arm hard/cooperative brakes rose by an order of magnitude.
- **Penetration benefit is still weak/noisy.** Even post-fix, collisions rise from baseline to any connected condition then plateau; arms stay close. Single seed; Town10 dominates.
- **Immobilise-as-obstacle is real but secondary.** The primary/secondary split shows pileups into frozen wrecks are a minority on Town01/Town05, larger on Town10; the primary metric removes that confound from the headline.
- **Residual hypotheses** for the flat curve: cautious CAVs braking in hostile traffic invite rear-ends from non-connected HDVs; the Python brake override interacts with the Traffic Manager; 60 s single-seed cells have high variance.

9. Threats to validity

- Single seed (seed 0): no real variance bars; directional only.
- Single 60 s duration; single seed; metric coupling from immobilise-in-place (mitigated by the primary metric, not eliminated).
- Run completeness: 72/72 with one auto-recovered retry.

10. Recommended next steps

- Re-run as Configuration B (3 seeds, 120 s) for real variance bars and a statistically meaningful penetration curve.
- Add a follower-aware / graduated CAV brake policy so emergency stops do not invite rear-ends; audit the Python-override vs Traffic-Manager interaction.
- Keep out/ablation_G (v1) and out/ablation_G2 (v2) archived; this report supersedes the v1 report for the corrected numbers.

Data source: out/ablation_G2 (v2, post-fix); comparison baseline out/ablation_G (v1). Generated by scripts/make_ablation_G2_report.py directly from per-cell JSON metrics.